

IOT Based **Pet Safety Collar**

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Introduction

What is IoT based Pet Safety Collar?

It is a smart pet collar equipped with sensors and connectivity to monitor pet's health and location in real-time. It also has a feature that activates the led light in the collar when it detects vehicle within a certain proximity range alerting the driver of the presence of the pet.

The main aim of this project is to protect pets and any other animals from serious accidents on the road.



Problems Explanation

Pets can go missing due to carelessness of the owners.

Many animals, especially pets cannot be seen by unconscious drives at night.

Pet's health issues are often unnoticed.

Existing products are either expensive or have limited features.



Project Concept

Use the Pet Safety Collar to collect and send data.
Monitor pet's heart rate, oxygen levels, temperature and location.
Send alerts for location change, health status and upcoming vaccinations.

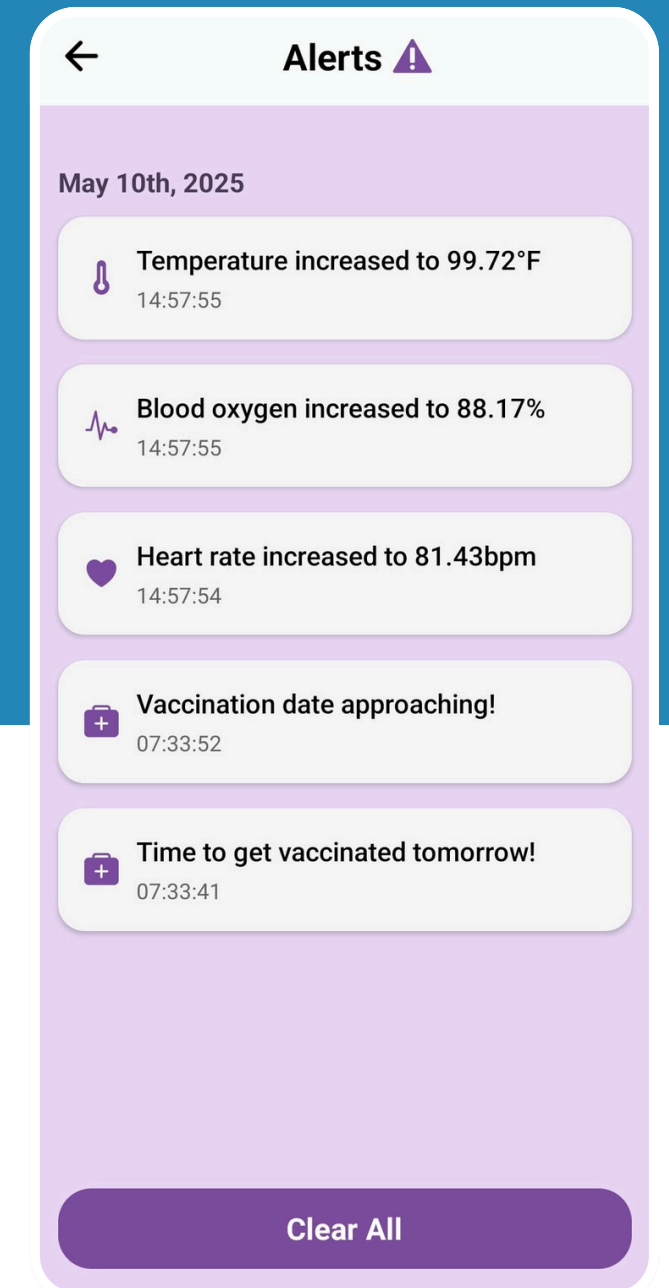
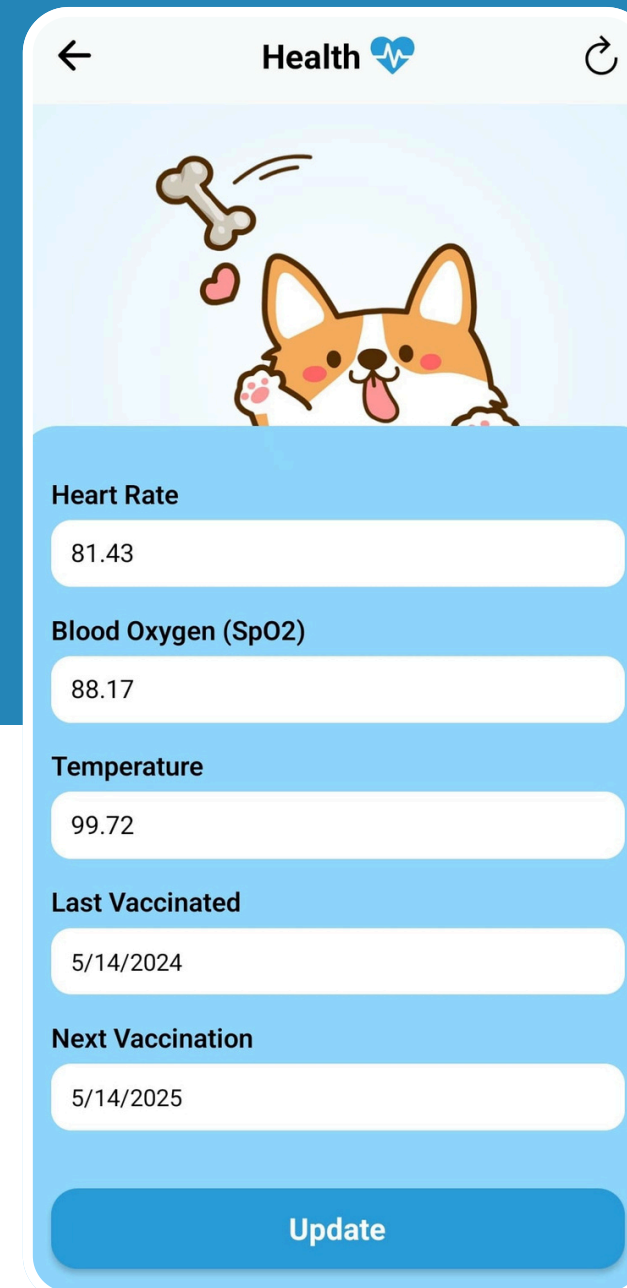
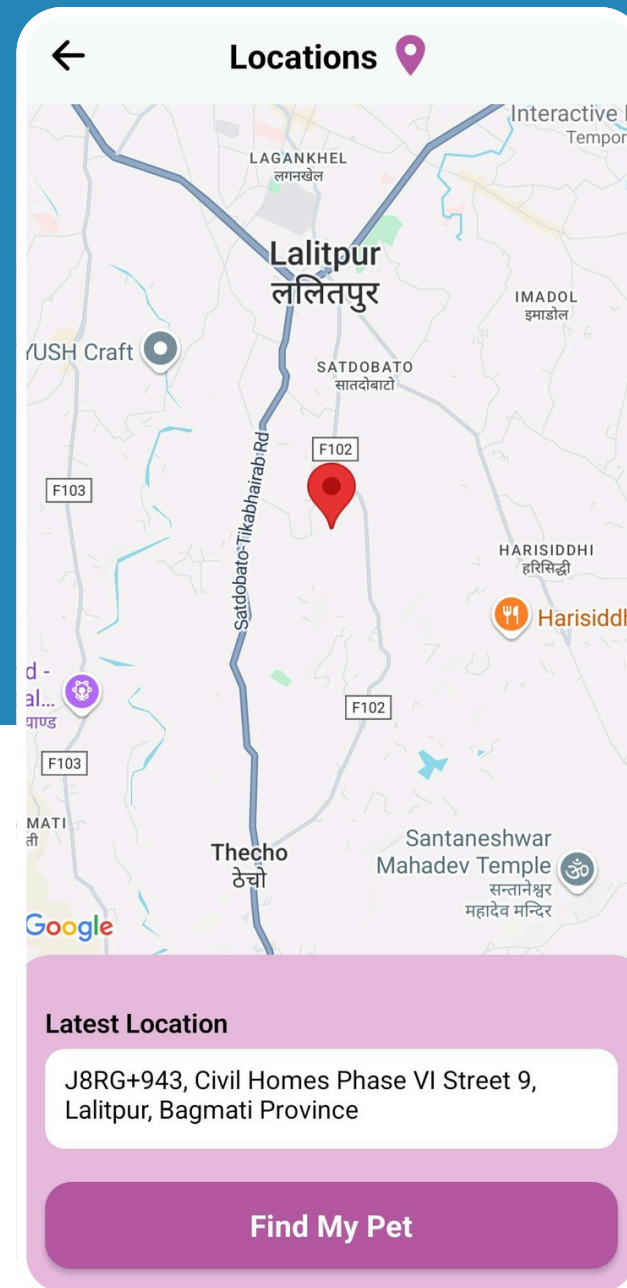
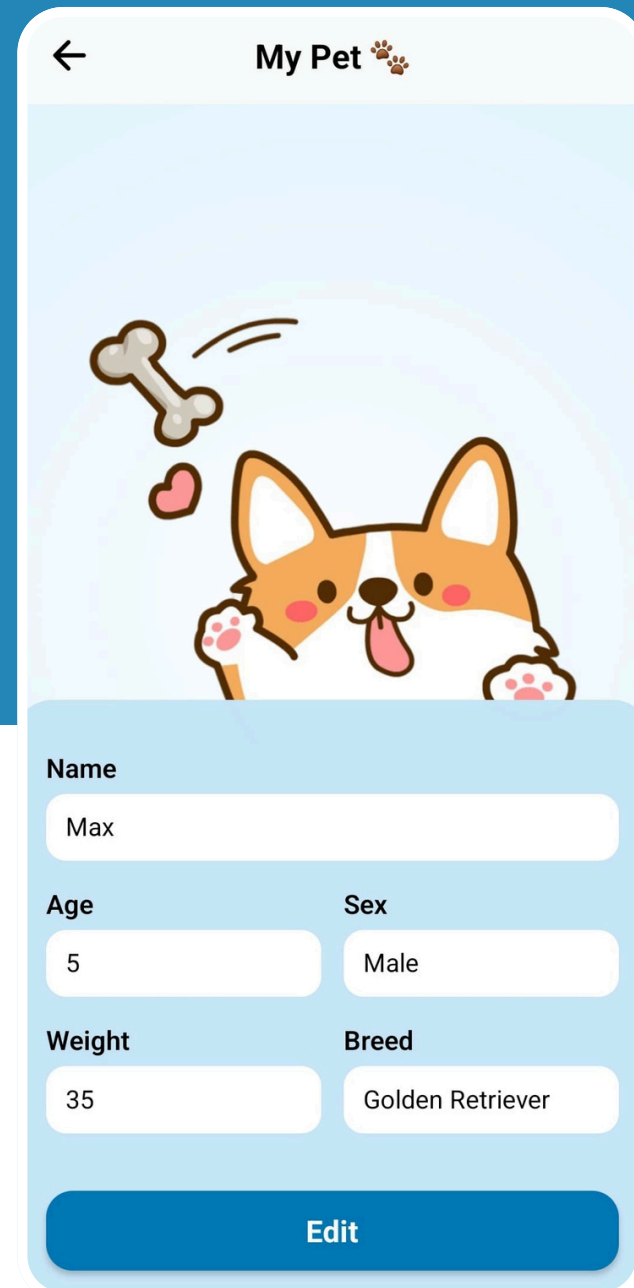
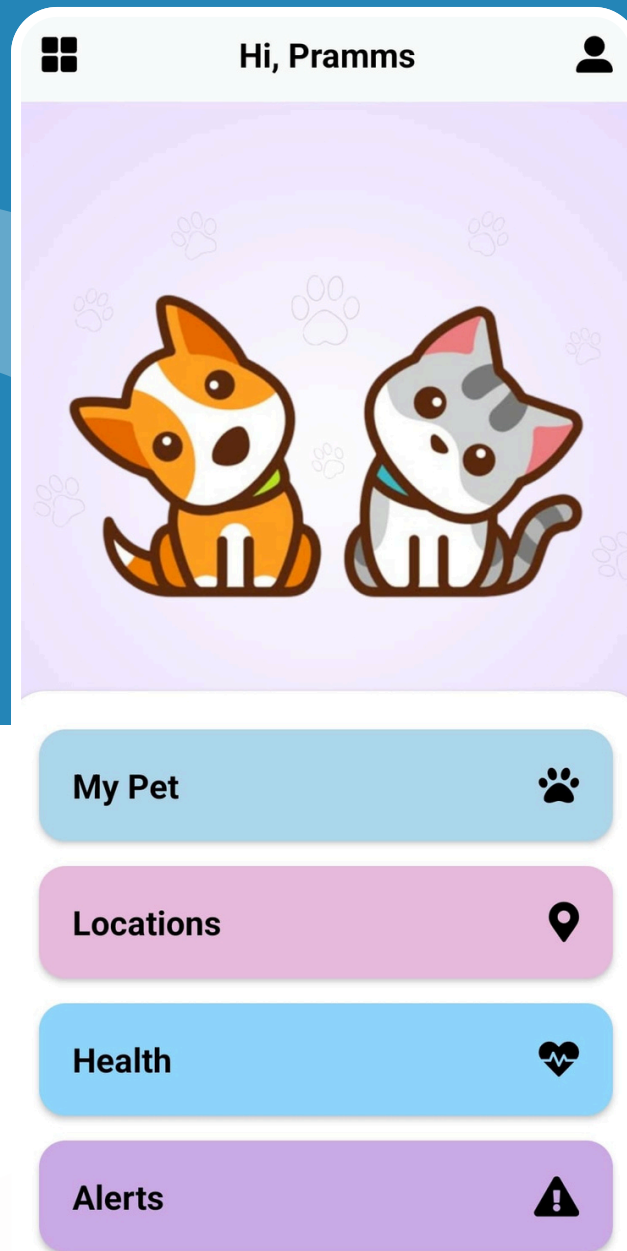
System Architecture

Hardware	Software
ESP32-C3 Super Mini	React-native
Ultrasonic Sensor HCSR04	Firebase
GPS Module NEO-6M	Arduino IDE
MAX30100 Pulse Sensor	Expo
Led light	

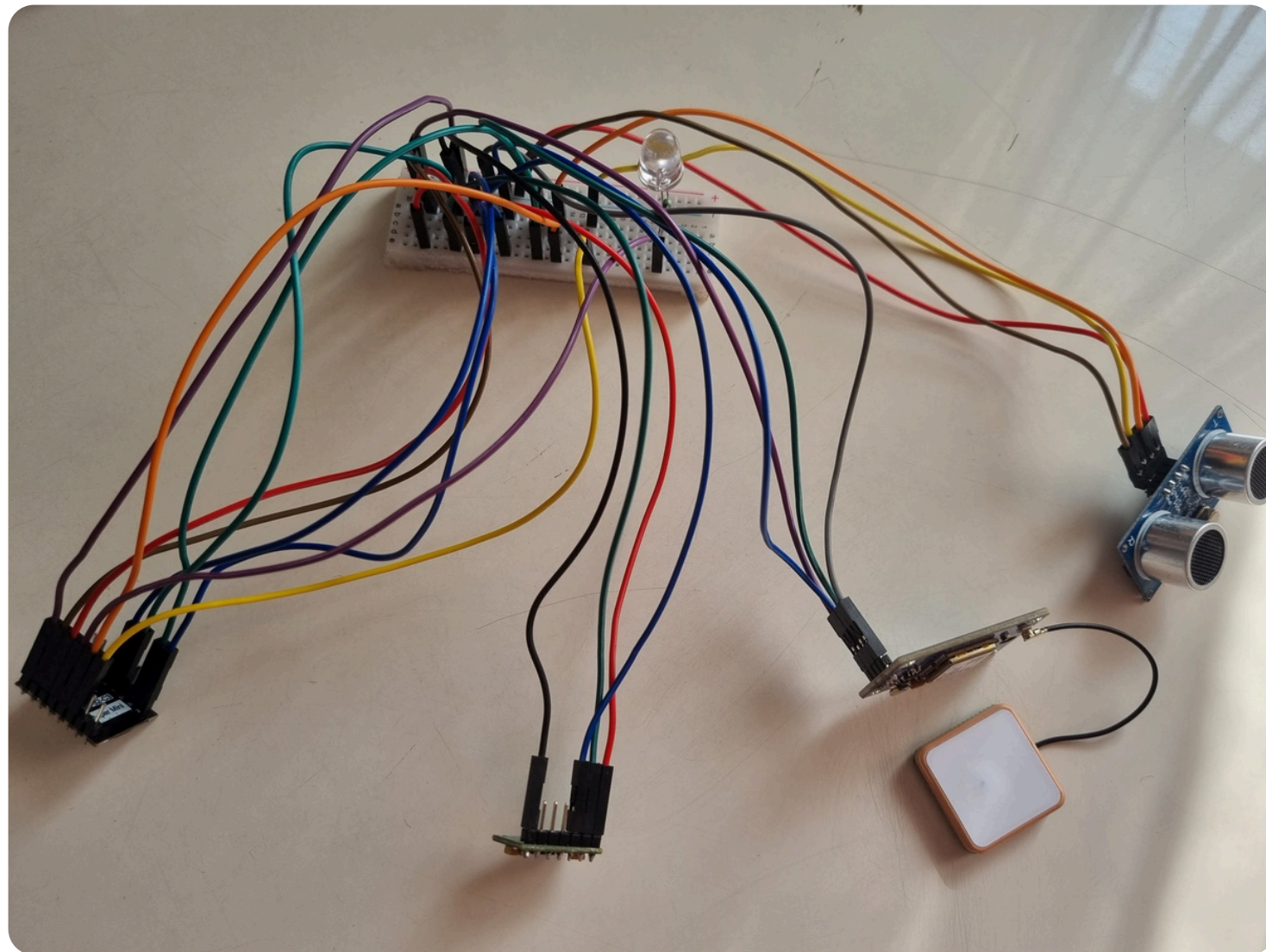
Implementation Overview

- The ESP32-C3 reads the data for the GPS and the pulse sensor.
- Then the data is sent to firebase using REST API.
- The mobile app then fetches and displays this data.
- Finally the alerts are triggered based on the location change, high or low heart rates, oxygen levels, temperature and vaccination updates.

App Features



Collar Design



Key Benefits

- Alert drivers of the pet's presence using led.
- Real-time monitoring of pet safety.
- Early detection of health issues.
- GPS tracking of pet's location.
- Cost-effective and expandable system.
- Peace of mind for owners.



Thank You

